

CampusEats Orders REST Service

CS543 · Web Services · Assignment 5: HTTP Methods & Headers

Group ID: 12

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Implementation Summary

The CampusEats service demonstrates HTTP method semantics, request and response headers, authentication, CORS, rate limiting, ETags, conditional requests, idempotency keys, and retry-safe write behaviour.

Method Map

Action	Method	Endpoint	Safe	Idempotent
Create order	POST	/orders	No	Via Idempotency-Key
List orders	GET	/orders?status=...	Yes	Yes
Read order	GET	/orders/{id}	Yes	Yes
Modify address	PATCH	/orders/{id}	No	If-Match protected
Cancel order	POST	/orders/{id}/cancel	No	State controlled
Delete order	DELETE	/orders/{id}	No	Yes
Capability discovery	OPTIONS	/orders	Yes	Yes

Header Coverage

Content-Type and Accept negotiation; Authorization Bearer headers; ETag, If-None-Match and If-Match; Idempotency-Key; rate-limit headers; CORS; X-Content-Type-Options; and Strict-Transport-Security.

Verification

The repository contains automated tests, OpenAPI validation, and a curl transcript covering successful creation, an idempotent repeat, conditional GET behaviour, conditional write protection, and representative failure paths.